

THAYER SCHOOL OF
ENGINEERING
AT DARTMOUTH

Death on Two Wheels

A Statistical Investigation of Motorcycle Risks and Public Perception

William Cast

ENGS 93 with Dr. Ronald Lasky

Spring 2023

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1 – Introduction

A personal note: I have loved everything motorized for as long as I can remember. This passion has led me to work on RC planes, Go-Karts, boats, and cars. With each of these projects, I have received nothing but support from my parents until I arrived home with a 1966 90cc motorcycle in the back of a pickup truck. Of the many parents I have spoken with about this, only two have supported my working on and riding this motorcycle, so a question formed: **What are the perceived risks versus the actual dangers of riding a motorbike compared to other forms of transportation, and what influences public opinion?**

To answer this question, we will first examine media coverage of motorcycling to determine if the media positively or negatively represents motorcycling. We can then survey the public and assess whether the media has statistically impacted public opinion on motorcycle safety. Then we will compare actual crash statistics to public opinion and see how they correlate. Next we will complete a safety report using the data collected by the Insurance Institute for Highway Safety (IIHS), the National Highway Traffic Safety Administration (NHTSA), and the Bureau of Transportation Statistics (BTS). Combining these details will expose the significance of the impact the media and other sources might have on the public opinion of motorcycle safety, as well as helping us determine which motorcycle habits are negatively affecting actual motorcycling safety.

2 – The Data

2.1 – Media Coverage

Media coverage biases the public's view of activities through different methods, but one of the most significant is through media amplification bias. This is a phenomenon in which media coverage disproportionately amplifies certain aspects. In this case, I hypothesize that media coverage disproportionately highlights the risks of motorcycling - crashing, running from police, speeding - which activates the availability heuristic. This means that when people evaluate motorcycle safety, more memorable and dangerous extremes come to mind, leading to an inflated perception of risk.

To test this, we first have to look at reports from news sites that mention “motorcycle.” We will look at several sources to eliminate as much bias as possible, including the Seattle Times, New York Times, Fox News, Fox 59, East Idaho News, and the top 50 grossing movies of all time. There are sources of different sizes, political views, and media types across the US. Doing much of the research myself means I can only look through so much media, so there will inevitably be oversights (See section 5.1), but I do my best to minimize those. Of the 106 Articles/Reports/Movies that include motorcycles from the last 30 days, 52% portrayed motorcycles and motorcyclists in a bad light. Of the negative stories, 51% spoke specifically about motorcycle crashes and deaths, 23% of the adverse reports told of police shootouts with motorcyclists or gang-related motorcycle drive-by shootings. Only 15% of all publications positively portray motorcyclists, whether it was a motorcycle rally for a good cause or a wholesome story of newlyweds motorcycling across the US. 34% of the articles that mentioned the word “Motorcycle” neither praised nor criticized, simply discussing new tariffs or mentioning a motorcycle to no effect. It appeared local news sources were significantly more likely to report crashes and biker deaths than nationwide news sources. To test this, I conducted a Z-Test where:

H_0: $P_l = P_n$ — the probability of a local source reporting death/injury is the same as a national one.

H_2: $P_l > P_n$ — the probability of a local news source reporting death/injury is higher.

Where $p = \frac{x_l + x_n}{n_l + n_n}$, $p_l = \frac{x_l}{n_l}$, $p_n = \frac{x_n}{n_n}$, and $z = \frac{p_l - p_n}{\sqrt{p(1-p)(\frac{1}{n_l} + \frac{1}{n_n})}}$. This gives me a Z-score of 3.32, which gives a one-tailed p-value of 0.00045. I can then calculate the confidence interval using $SE = \sqrt{\frac{p_l(1-p_l)}{n_l} + \frac{p_n(1-p_n)}{n_n}}$, and the two-tailed Z-score for 95%, which is 1.96, to calculate my confidence interval. $CI = (p_l - p_n) \pm z \cdot SE$. This gives me a 95% confidence interval that local news sources are between 12% and 46.6% more likely to report motorcycle-related crashes and deaths than larger media platforms.

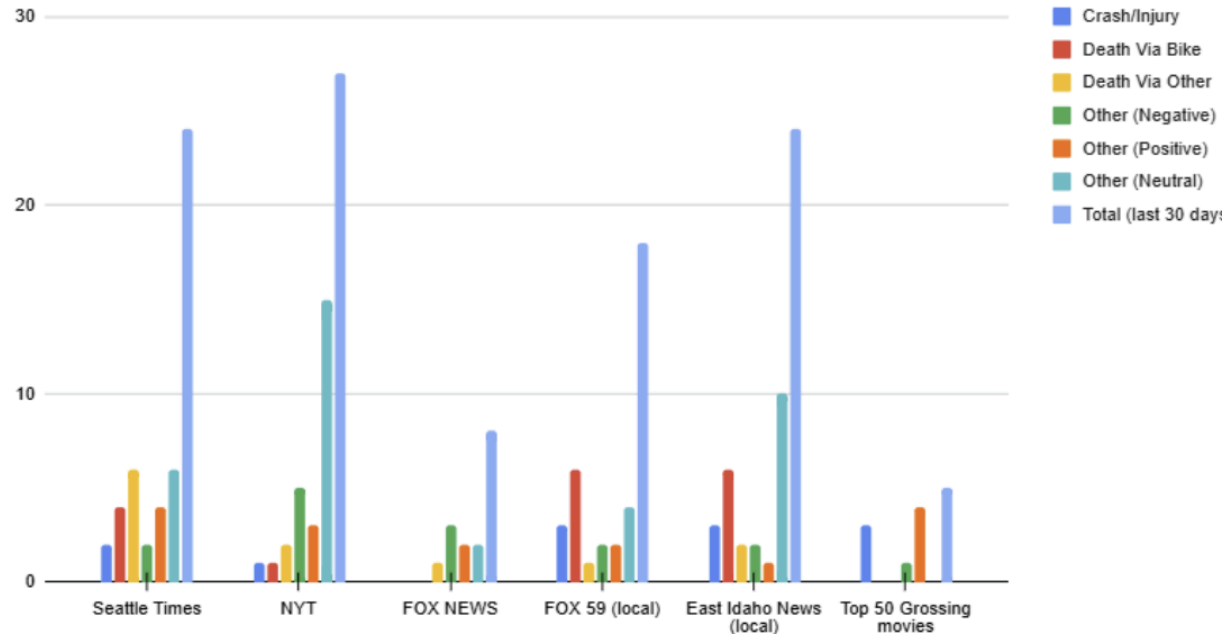
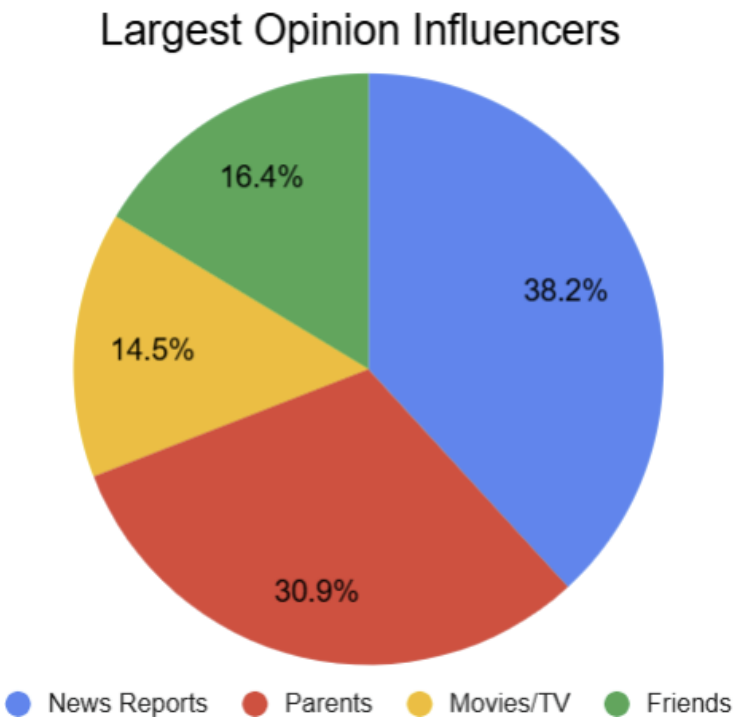


Figure 2.1 — Media Coverage
This chart shows the topics of articles posted in the last 30 days on different news forums. Each article is only counted in one category. Crash/Injury represents the crashes or injuries discussed in articles/movies (not including deaths). Death Via Bike represents the media in which a person riding or struck by a bike is killed. Death Via Other counts media in which people die, not explicitly related to the motorcycle mentioned (multiple counts of gang violence and police shootings). Other (Negative) articles cast a negative light on motorcycles. Other (Positive) media cast a positive light on Motorcycles. Other (Neutral) articles mention motorcycles neither in a positive nor negative light. Total counts the total media from each source that contains the word motorcycle. The top 50 grossing movies ever are used instead of the last 30 produced, and those that have scenes where a motorbike is displayed for more than 10 seconds are included rather than the word “motorcycle.”

2.2 – Public Opinions

There are no accessible statistics on what sources have had the most significant impact on public opinion of motorcycle safety, so I gathered data by launching a survey. I surveyed only three students on campus before posting my survey online to minimize the bias Dartmouth introduces. I had hoped to get close to 100 responses, but ended up with 31. This data was crucial to determining the effect the media and other sources had on the public and its severity.

My research showed that of the 31 respondents, only 10 would ride a motorcycle for any distance. A fewer 5 believe it is a safe form of transportation. However, 7 people deemed it a safe form of transportation if "Only used rarely." So, 13 of 31 respondents said that it is safe to some level. 10 responded that they would ride a motorcycle for some time, yet there was no significant overlap between those who would ride a bike and those who found it safe. Using the formula $Overlap = \frac{|A \cap B|}{|A \cup B|}$, we find that 30% of those who said they would ride a motorcycle do not believe it is safe to do so for any distance. Similarly, 46% of those who deemed it safe to some level responded that they would still not ride for any distance. With my media and public research complete, I needed to begin collecting data on safety statistics from crash reports around the US.



2.3 – Safety Statistics

I began collecting actual crash data from several sources, including InjuryFacts.org, the National Highway and Traffic Safety Administration (NHTSA), and the Insurance Institute for Highway Safety (IIHS). It is generally accepted that the risk of injury or death on a motorcycle is roughly 20-30 times greater than that of a car per mile. Still, this data will allow me to check that statistic, generate relevant facts, and compare them with public beliefs. This data was compiled in different ways in Google Sheets and JMP, and is available in Appendix 2.3. Before going into analysis, the key comparisons include deaths per 100 million vehicle miles (MVM) traveled, which for motorcyclists was 31.39 compared to 1.33 deaths per 100 MVM for cars, meaning that per mile driven, the biker is taking 23.6 times the risk of the car driver. The car crash rate was roughly 0.002 per 1000 miles driven compared to 0.00507 per 1000 miles for motorcycles, meaning motorcycles are 2.535 times more likely to crash than car drivers (NHTSA).

3 – Analyzing the Data

3.1 – Public Influence

To determine whether media influences public perception of how safe motorcycling is, I ran a logistic regression to predict whether individuals deemed motorcycling safe using perceived negativity of media and whether respondents reported news or movies/TV were major influencers on their opinion.

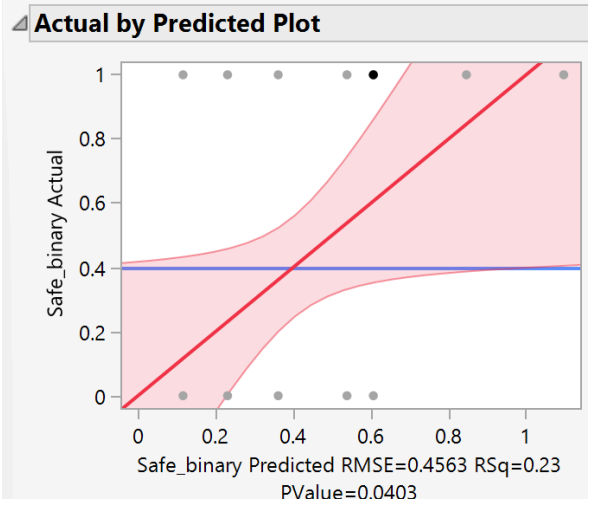
My null hypothesis states that the interaction between media influence and media negativity does not affect perceived safety.

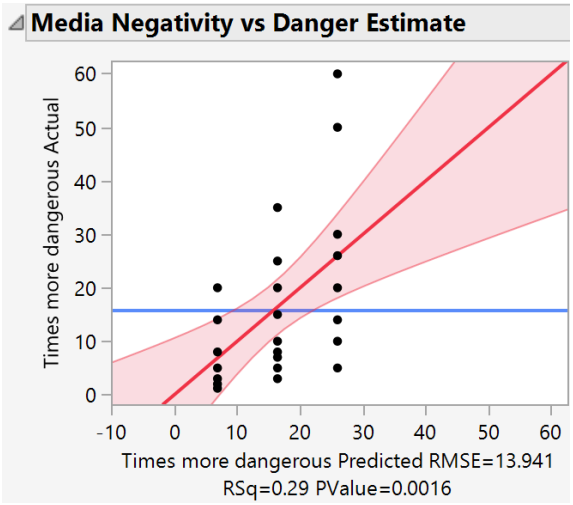
H_0: $\beta_{interaction} = 0$

My alternative hypothesis stated that media influence, media negativity, or their interaction had some effect on perceived safety.

H_1: $\beta_{interaction} \neq 0$

I used the equation $log(\frac{p}{1-p}) = \beta 0_0 + \beta_1 X1_1 + \beta_2 X_2 + \ldots$ and found that alone, media influence gave me P=0.4354 and media negativity gave me a P=0.1836. However, by multiplying them and including the initial variables, I get P=0.0403. This means there is greater than 95% confidence that the combination of media influence and perceived media negativity has a statistically significant adverse effect on how safe a person perceives motorcycling. Our pseudo-R^2 of 0.23 does not seem meaningful, but it is statistically significant given that we are modeling human perception with just two variables. I then ran a linear regression to determine the impact of perceived media negativity on estimated danger when compared to cars. With a P-value of 0.0016, I can confidently say media negativity significantly impacts perceived safety. We can use the equation:





Times_More_Dangerous ≈ -21.7606 + 9.54327(Percieved_Media_Negativity) to predict the respondents' estimated times more dangerous given their perception of the media's influence. This allows us to predict the respondent's guess of safety levels with a standard error of 11.07 for the intercept and 9.543 for the slope.

Next, I wanted to compare the estimated danger rating to actual statistics. In 2023, 31.39 motorcyclists died per 100 million vehicle miles compared to 1.38 per 100 million car miles driven. This equates to riding a motorcycle being 23.6 times more dangerous than a car per mile (See Appendix 2.3). My survey reveals that people believed motorcycling to be 15.797 times more hazardous than driving a car. JMP gives a standard deviation of 16.32 with a standard error of the mean of 2.93. This is significant because the upper 95% mean is 21.784, meaning people significantly underestimate the actual danger of motorcycling.

3.2 – Lowering Risk

People significantly underestimate the danger of riding a motorcycle despite the substantial negative impact of the media. I asked two friends who own and ride motorcycles what they thought about this and the increased risk of motorcycle riding. The response I received indicated they believed there was a Type II error in the statistics presented. Given the lack of comprehensive surveying, this is a difficult thing to test, so we have to make some assumptions. For example, from InjuryFacts.com, we know 25% of riders admitted to drinking and riding occasionally. That 25% causes 100% of drunk riding crashes as well as 25% of other crashes (when they aren't drinking), meaning their relative risk is 2.08 times that of somebody who never drinks and rides. I considered this for 6 categories: drinking, speeding, riding in bad conditions, not wearing a helmet, riding at night, and riding with cars around. Each with a respective risk modifier of 2.08, 1.5, 1.24, 1.37, 1.639, and 1.37. These values are determined by looking at the different statistics across the IIHS, NHTSA, and Injury Facts websites and compiling the crashes to determine the increased risk per habit. From this, I formulated a program[1] with significant online help that lets you test different hypothetical riding styles and compare the risk to other riders and the risk of driving a car.

4 – Conclusion

The negative public perception of motorcycle safety is primarily due to the publications of different media types, with less than 3% of my survey respondents noting positive motorcycle implications in publications. This matched my findings in recent news reports and top movies. 53.0% of the publications I reviewed put motorcycles and their users in a negative light. 27.4% of the reviewed material specifically talked about severe injuries or deaths of those on motorcycles. Only 15.1% of the publications positively portrayed motorcyclists. Despite the negative outlook created by the media, survey participants significantly underestimated the risk of motorcycling compared to driving a car. Without consideration, riding a motorcycle is roughly 23.6 times more dangerous than driving a car. This means that every 43.96 miles ridden carries the same risk as driving your car 1000 miles. Wearing a helmet, only riding in good conditions during the day, avoiding congested areas, and not drinking or speeding reduces your risk to only a few times that of the average car driver. This is not to say that motorcycling is safe. It is not. However, significant steps should be taken to minimize their risk.

5 – Next Steps

5.1 – Oversights

As with any study, some oversights are made. Due to the scale of my research, there are a few significant sources of error. With my media research, I could only check through 6 different sources and only the last 30 days (or 50 top movies) worth of publications. Similarly, with each publication review, it was up to my discretion whether or not the report favored, disfavored, or imparted no bias on motorcycles and their riders. Despite my attempt to be unbiased, others would likely have different opinions on some of the publications.

My research on public opinions consisted of creating a Google form and posting it online. This comes with some unavoidable problems. Many of the respondents were in their 10s and 20s

and, therefore, likely more influenced by their parents than older people would have been. ¾ of respondents over 39 did not report that their parents significantly impacted their opinions. This differs from the 54.8% who said their parents did influence their views. We have nearly balanced gender responses with 16 coming from women, 14 from men, and one from others. Of the respondents, all but two seem to have given genuine responses. The two were distinguishable because one reported being older than 99 and the other, when asked for a number, responded “the car is dangerouser.”

Finally, I must admit to some bias; despite trying to eliminate all bias from this research, I have always wanted a motorcycle. I would have loved to find out that their danger is overstated. My research has shown that it isn't the case, but it must be said.

5.2 – Further Research

The key areas for future research include broadening the range of news and media sources examined and increasing the number of respondents for the survey. To make people more willing and honest in the survey, I kept it relatively short, but having more data would also be extremely helpful. It would also be beneficial to have better data on motorcyclists' habits themselves, because though much of what I needed was available, it was challenging to track down.

6 – References

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Rowe, Robert J. *Contemporary Attitudes Toward Motorcycle Riding Safety and Riding Risk Factors – Part 2*. International Motorcycle Safety Conference, 30 Mar. 2006, Irwin Broh & Associates.

"1983 Motorcycle Tariff." *Wikipedia*, Wikimedia Foundation, 2024, https://en.wikipedia.org/wiki/1983_motorcycle_tariff Accessed 18 May 2025.

7 – Appendices

Coverage of	Seattle Times	NYT	FOX NEWS	FOX 59 (local)	East Idaho News	Top 50 Grossing movies
Crash/Injury	2	1	0	3	3	3
Death Via Bike	4	1	0	6	6	0
Death Via Other	6	2	1	1	2	0
Other (Negative)	2	5	3	2	2	1
Other (Positive)	4	3	2	2	1	4
Other (Neutral)	6	15	2	4	10	0
Total (last 30 day	24	27	8	18	24	5

Appendix 2.1 - Portrayal of Motorcycling in Common Media Forums

Timestamp	Would you consi	What/Who has influenced your	Do you believe riding a r	How many times more dangerous	How old are you (1 star = 10 - 19, 2	Gender	How has mec	Where do you think most motor
5/20/2025 0:48:24	Yes	Friends	Only if used rarely	8	2	Female	4	Collision With other vehicle
5/20/2025 0:48:51	Yes	Parents, Movies/TV	Yes	5	2	Female	3	Collision With other vehicle
5/20/2025 0:52:10	Yes	Parents, Friends	Only if used rarely	10	1	Male	5	With no other vehicle
5/20/2025 1:09:23	No	Movies/TV, News Reports	No	35	2	Male	4	Collision With other vehicle
5/20/2025 1:09:58	No	Parents, News Reports	No	20	3	Male	5	Collision With other vehicle
5/20/2025 1:10:44	No	Movies/TV, News Reports	No	30	4	Female	5	Collision With other vehicle
5/20/2025 1:11:24	Yes	Parents	No	25	2	Male	4	With no other vehicle
5/20/2025 1:49:47	No	Parents, News Reports	No	5	2	Male	3	With no other vehicle
5/20/2025 4:03:25	Yes	Friends, News Reports	Yes	14	3	Other	3	Collision With other vehicle
5/20/2025 4:57:32	No	Friends, News Reports	Yes	3	2	Female	3	With no other vehicle
5/20/2025 5:47:37	No	News Reports	Only if used rarely	10	2	Female	4	Collision With other vehicle
5/20/2025 10:12:36	Yes	Movies/TV	Yes	3	1	Male	4	Collision With other vehicle
5/20/2025 13:02:12	No	Parents, News Reports	No	5	2	Male	4	With no other vehicle
5/20/2025 13:11:04	No	Parents, Movies/TV	Only if used rarely	5	1	Male	4	With no other vehicle
5/20/2025 14:02:07	No	Friends	No	20	2	Male	4	Collision With other vehicle
5/20/2025 17:43:26	No	Parents, News Reports	No	7	2	Male	4	Collision With other vehicle
5/20/2025 17:47:10	Yes	Friends, Movies/TV	No	5	4	Male	4	Collision With other vehicle
5/20/2025 20:36:29	No	Parents	No	8	3	Female	3	With no other vehicle
5/20/2025 23:55:27	No	Parents, News Reports	Only if used rarely	14	6	Male	5	Collision With other vehicle
5/21/2025 2:09:08	No	Parents, Movies/TV, News Reports	No	5	1	Female	5	With no other vehicle
5/21/2025 2:28:26	No	Parents, News Reports	No	3	2	Female	4	Collision With other vehicle
5/21/2025 3:28:09	No	Parents, News Reports	No	20	2	Female	3	With no other vehicle
5/21/2025 3:31:08	No	News Reports	No	60	2	Female	5	Collision With other vehicle
5/21/2025 3:31:34	No	Parents, News Reports	No	15	2	Female	4	With no other vehicle
5/21/2025 3:31:51	No	News Reports	No	60	2	Female	5	Collision With other vehicle
5/21/2025 3:32:06	No	News Reports	No	26	5	Male	5	Collision With other vehicle
5/21/2025 3:32:13	No	Parents, News Reports	Only if used rarely	50	2	Female	5	Collision With other vehicle
5/21/2025 3:37:47	Yes	Parents, Friends	Yes	1.2	2	Male	3	Collision With other vehicle
5/22/2025 22:50:35	No	Movies/TV, News Reports	Only if used rarely	2	1	Female	3	Collision With other vehicle
5/22/2025 23:02:34	Yes	Friends, News Reports	Yes	0.5	10	Female	1	Collision With other vehicle
5/22/2025 23:03:10	Yes	Parents, News Reports	No	15	6	Female	4	Collision With other vehicle

Appendix 2.2 - Responses to Public Opinion Survey

Motorcyclist deaths and death rates, 2007-2023

Year	Deaths	Death rate per 100 million vehicle miles traveled
2007	5,174	24.18
2008	5,312	25.52
2009	4,469	21.46
2010	4,518	24.40
2011	4,630	24.97
2012	4,986	23.41
2013	4,692	23.04
2014	4,594	23.00
2015	5,029	25.65
2016	5,337	26.09
2017	5,226	25.94
2018	5,038	25.09
2019	5,044	25.62
2020	5,620	31.31
2021	6,144	31.28
2022	6,251	26.30
2023	6,335	31.39

Source: NSC analysis of NHTSA FARS data and National Center for Statistics and Analysis. Motorcycles: 2017 data (Updated, Traffic Safety Facts. Report N. DOT HS 812 785). Washington, DC: National Highway Traffic Safety Administration.

	37.90%	36.46%
	1.34%	19.34%
	40,901	
	15.5%	

Appendix 2.31

Motorcyclist injuries and injury rates, 2007-2022

Year	Injuries	Injuries rate per 100 million vehicle miles traveled
2007	103,000	481
2008	96,000	461
2009	90,000	432
2010	82,000	443
2011	82,000	442
2012	93,000	437
2013	89,000	437
2014	92,000	461
2015	89,000	454
2016*	104,000	508
2017*	89,000	442
2018*	82,000	408
2019*	84,000	427
2020*	79,000	440
2021*	84,898	432
2022*	82,088	345
2023*	82,564	409

* Starting in 2016 NHTSA transitioned from NASS GES estimate system to CRSS. CRSS estimates and NASS GES estimates are not comparable due to different sample designs.

Source: NSC analysis of NHTSA CRSS data and National Center for Statistics and Analysis. (2019, July). Motorcycles: 2017 data (Updated, Traffic Safety Facts. Report N. DOT HS 812 785). Washington, DC: National Highway Traffic Safety Administration.

Appendix 2.32

Motorcycle traffic death overview, 2022

Factor	Number	Percent
Rural/ Urban		
Rural	2,223	35%
Urban	4,076	64%
Unknown	36	1%
Weather		
Clear/Cloudy	5,933	94%
Rain	129	2%
Fog	35	1%
Other/Unknown	238	4%
Time of Day		
Daytime	3,117	49%
Nighttime	3,184	50%
Unknown	34	1%
Crash Type		
Single-vehicle crash	2,279	36%
Two-vehicle crash	3,578	56%
Three or more vehicle crash	478	8%
Helmet Use		
Helmeted	3,949	62%
Unhelmeted	2,162	34%
Unknown	224	4%
Number of motorcycle operators by impairment		
Alcohol impaired (BAC=0.08+ g/dL)	1,668	26%
Not alcohol impaired	4,761	74%

NSC analysis of NHTSA FARS data

Appendix 2.33

Registered motorcycles and vehicle miles traveled, 2007-2023

Year	Registered vehicles	Vehicle miles traveled (millions)
2007	7,138,476	21,396
2008	7,752,926	20,811
2009	7,929,724	20,822
2010	8,009,503	18,513
2011	8,437,502	18,542
2012	8,454,939	21,298
2013	8,404,687	20,366
2014	8,417,718	19,970
2015	8,600,936	19,606
2016	8,679,380	20,455
2017	8,715,204	20,149
2018	8,666,185	20,076
2019	8,596,314	19,688
2020	8,347,435	17,947
2021	9,424,769	19,642
2022	9,186,256	23,765
2023	9,516,910	20,181

Source: Federal Highway Administration. Highway Statistics Series: https://www.fhwa.dot.gov/policyinformation/statistics.cfm

Appendix 2.34

Whole Model

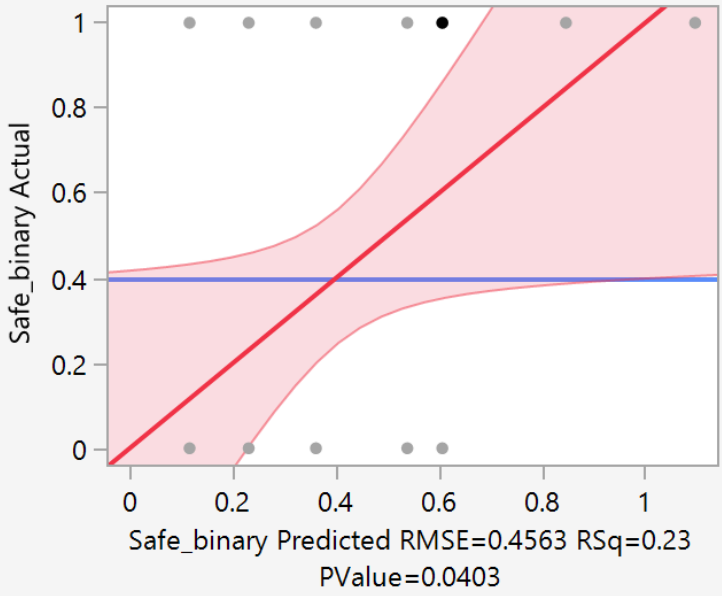
Effect Summary

Source	Logworth	PValue
Media_Influenced*Media Negative Score	1.676	0.02110
Media Negative Score	0.736	0.18357 ^
Media_Influenced	0.361	0.43540 ^

Remove Add Edit Exclude ☐ FDR

('^' denotes effects with containing effects above them)

Actual by Predicted Plot



Summary of Fit

RSquare	0.231596
RSquare Adj	0.157235
Root Mean Square Error	0.456303
Mean of Response	0.4
Observations (or Sum Wgts)	35

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	3	1.9454094	0.648470	3.1145
Error	31	6.4545906	0.208213	Prob > F
C. Total	34	8.4000000		0.0403*

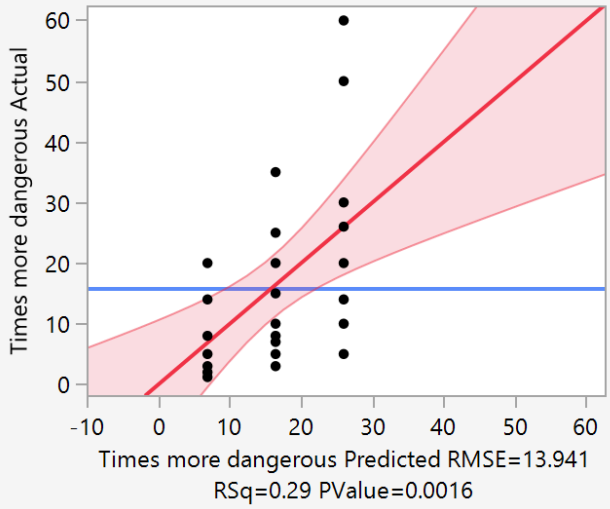
Appendix 3.11

Response Times more dangerous

Whole Model

Regression Plot

Media Negativity vs Danger Estimate



Summary of Fit

RSquare	0.294816
RSquare Adj	0.270499
Root Mean Square Error	13.94056
Mean of Response	15.79677
Observations (or Sum Wgts)	31

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	1	2356.1710	2356.17	12.1240
Error	29	5635.8387	194.34	Prob > F
C. Total	30	7992.0097		0.0016*

Lack Of Fit

Source	DF	Sum of Squares	Mean Square	F Ratio
Lack Of Fit	2	606.5814	303.291	1.6282
Pure Error	27	5029.2572	186.269	Prob > F
Total Error	29	5635.8387		0.2150
				Max RSq
				0.3707

Appendix3.12

	Not Drinking	Speeding	Bad conditions	helmeted	Nightime riding	Multi Vehicle
Percent of crash	0.64	0.36	0.06	0.62	0.5026	0.6403
Percent who Do	0.75	0.42	0.2	0.74	0.44	0.634
risk w/	0.52	0.6288	0.248	0.54	0.7215	0.86819
risk w/o	0.48	0.3712	75.2	0.46	0.2785	0.36
Relative risk	2.08	1.5	1.24	1.37	1.639	1.37

Appendix 3.2